

Terms, Factors, Coefficients, and Degree

Answer Key

Algebra 1

Quick Answers

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|--------------------------|-----------------------|
| 1. $x^2 + 5x$ | 6. $7x + 3$ |
| 2. $x^2 + 8x + 15$ | 7. $2x^2 + 5x - 12$ |
| 3. $4x^3 - x^2 + 6x - 9$ | 8. $3x^2(2x - 5)$ |
| 4. $(x + 3)(x + 4)$ | 9. $2x^5 + 7x^3 - 4x$ |
| 5. $6x + 15$ | 10. $x^2 + 8x + 12$ |
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1. Area model for width x , length $x + 5$.

Solution: The two cells are $x \cdot x = x^2$ and $x \cdot 5 = 5x$, so the area is the sum of those two cells: $x^2 + 5x$

Reading the answer: two terms. The coefficient of x is 5; the coefficient of x^2 is 1 (an invisible 1 is still a coefficient). Degree 2, because the largest exponent among the terms is 2.

2. Model for $(x + 3)(x + 5)$.

Solution: cells x^2 , $5x$, $3x$, 15 . The two middle cells are like terms ($5x + 3x = 8x$), so four cells collapse to three terms: $x^2 + 8x + 15$

Why fewer terms: $5x$ and $3x$ have the same variable to the same power, so they combine. Cells count multiplications; terms count what is left after combining like terms.

3. Terms out of order: -9 , $6x$, $4x^3$, $-x^2$.

Solution: coefficient and degree of each term — -9 : coefficient -9 , degree 0 (a constant is $-9x^0$); $6x$: coefficient 6, degree 1; $4x^3$: coefficient 4, degree 3; $-x^2$: coefficient -1 , degree 2.

Standard form orders the terms by decreasing degree: $4x^3 - x^2 + 6x - 9$

The polynomial has degree 3 (the largest term degree) and leading coefficient 4 (the coefficient of the highest-degree term, not the number written first in the original list).

4. Factor $x^2 + 7x + 12$.

Solution: find two numbers with product 12 and sum 7: 3 and 4. $(x + 3)(x + 4)$

Degrees: each factor has degree 1, and the product has degree $1 + 1 = 2$ — multiplying polynomials adds their degrees.

5. $3(2x + 5)$ — sum of terms, or product of factors?

Solution: Ann is describing it as written: $3(2x + 5)$ is a single term built as a product of the factors 3 and $(2x + 5)$. Distributing rewrites it as a sum:

$$3(2x + 5) = 3 \cdot 2x + 3 \cdot 5 = 6x + 15$$

$$6x + 15$$

Factored form: factors 3 and $(2x + 5)$; one term. Expanded form: terms $6x$ and 15 ; no common factor written out. Jo is right about the expanded form only.

6. Add $(3x^2 + 5x - 1) + (-3x^2 + 2x + 4)$.

Solution: combine like terms by degree: $3x^2 + (-3x^2) = 0$, $5x + 2x = 7x$, $-1 + 4 = 3$.

$$\boxed{7x + 3}$$

What happened to the degree: the two degree-2 terms are opposites, so the x^2 term has coefficient 0 and disappears. The sum has degree 1. Adding polynomials can lower the degree; it can never raise it.

7. Model for $(2x - 3)(x + 4)$.

Solution: cells $2x \cdot x = 2x^2$, $2x \cdot 4 = 8x$, $-3 \cdot x = -3x$, $-3 \cdot 4 = -12$. Combine $8x + (-3x) = 5x$: $\boxed{2x^2 + 5x - 12}$

The circled cell is the bottom-right one, $-3 \cdot 4 = -12$: the constant term always comes from multiplying the two constant parts. The coefficient of x is 5.

8. Factor $6x^3 - 15x^2$ completely.

Solution: the greatest common factor of $6x^3$ and $15x^2$ is $3x^2$ (3 divides 6 and 15; x^2 is the highest power in both):

$$6x^3 - 15x^2 = 3x^2(2x - 5)$$

$$\boxed{3x^2(2x - 5)}$$

Factors: 3 (degree 0), x^2 (degree 2), and $(2x - 5)$ (degree 1). Degrees add: $0 + 2 + 1 = 3$, matching the degree of the original polynomial.

9. Multiply $(2x^3 - x)(x^2 + 4)$.

Solution:

$$\begin{aligned} (2x^3 - x)(x^2 + 4) &= 2x^3 \cdot x^2 + 2x^3 \cdot 4 - x \cdot x^2 - x \cdot 4 \\ &= 2x^5 + 8x^3 - x^3 - 4x \end{aligned}$$

Combine the like x^3 terms: $8x^3 - x^3 = 7x^3$. $\boxed{2x^5 + 7x^3 - 4x}$

Degree and leading coefficient: degree 5, leading coefficient 2. Predictable without multiplying everything: the highest-degree term of the product is the product of the two highest-degree terms, $2x^3 \cdot x^2 = 2x^5$, so the degree is $3 + 2 = 5$ and the leading coefficient is $2 \cdot 1 = 2$.

10. Is $x^2 + 8x + 12$ equivalent to $(x + 2)(x + 6)$?

Solution: expand the product — two numbers with product 12 and sum 8 are 2 and 6:

$$(x + 2)(x + 6) = x^2 + 6x + 2x + 12 = x^2 + 8x + 12$$

$$\boxed{x^2 + 8x + 12} \quad \text{so the two expressions ARE equivalent.}$$

Explanation (accept any equivalent reasoning; read the student's words). “Sum of three terms” and “product of two factors” describe two *forms* of writing, not two different expressions. The same number comes out for every value of x , so it is one expression with two useful forms: the factored form shows the zeros ($x = -2$ and $x = -6$), the expanded form shows the degree, the leading coefficient, and the constant term. Maya is confusing how an expression is written with what it is.